

System Trimming Report: magnetic_actuator_trim

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System Information

Number of States	3
Number of Inputs	1
State Variables	z, v, i
Input Variables	u
Operating Conditions	$\{ 'z': 0.005 \}$
Parameters	$\{ 'm': 0.05, 'g': 9.81, 'L': 0.01, 'R': 1.0, 'k': 0.0001 \}$

Equilibrium (Trim Point)

State Equilibrium (x_e):

z	0.005000
v	0.000000
i	2.242237

Input Equilibrium (u_e):

u	2.242237
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Linearized Model

A matrix (State Matrix)

0.0000	1.0000	0.0000
95.7082	0.0000	-8.7502
0.0000	-21.8755	-100.0000

B matrix (Input Matrix)

0.0000
0.0000
100.0000

C matrix (Output Matrix)

1.0000	0.0000	0.0000
0.0000	1.0000	0.0000
0.0000	0.0000	1.0000

D matrix (Feedthrough Matrix)

0.0000
0.0000
0.0000

Stability Analysis

Classification: unstable

Poles / Eigenvalues:

Pole	Value	λ
Pole 1	10.6859	10.6859
Pole 2	-8.7899	8.7899
Pole 3	-101.8960	101.8960

Diagnostics

Converged	Yes
Feasible	Yes
Timestamp	2026-02-09T09:36:39.210193